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BSCY-6

Digital Forensics

Assignment no 1

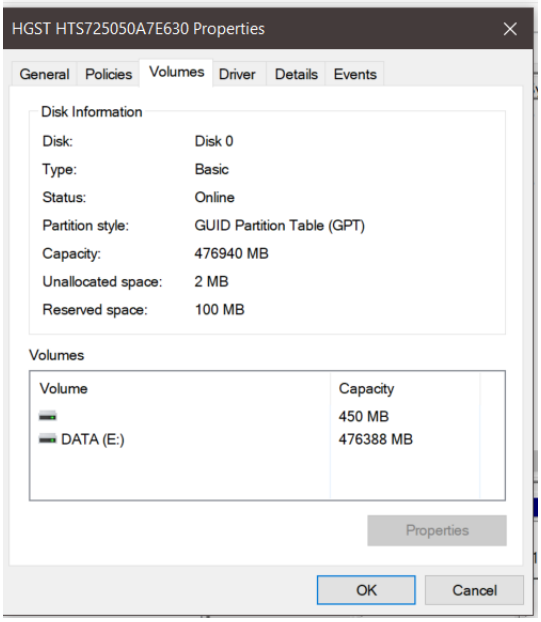
Forensic Analysis: Manual Partition Table Verification

1. Introduction

The objective of this assignment is to manually verify the size of a disk partition by analyzing the hex code of the partition table. While the assignment hint references MBR, my system utilizes the modern **GUID Partition Table (GPT)** standard. Therefore, this analysis focuses on extracting data from the GPT structures located at Sector 1 and Sector 2 of the physical drive.

System Information & Tools

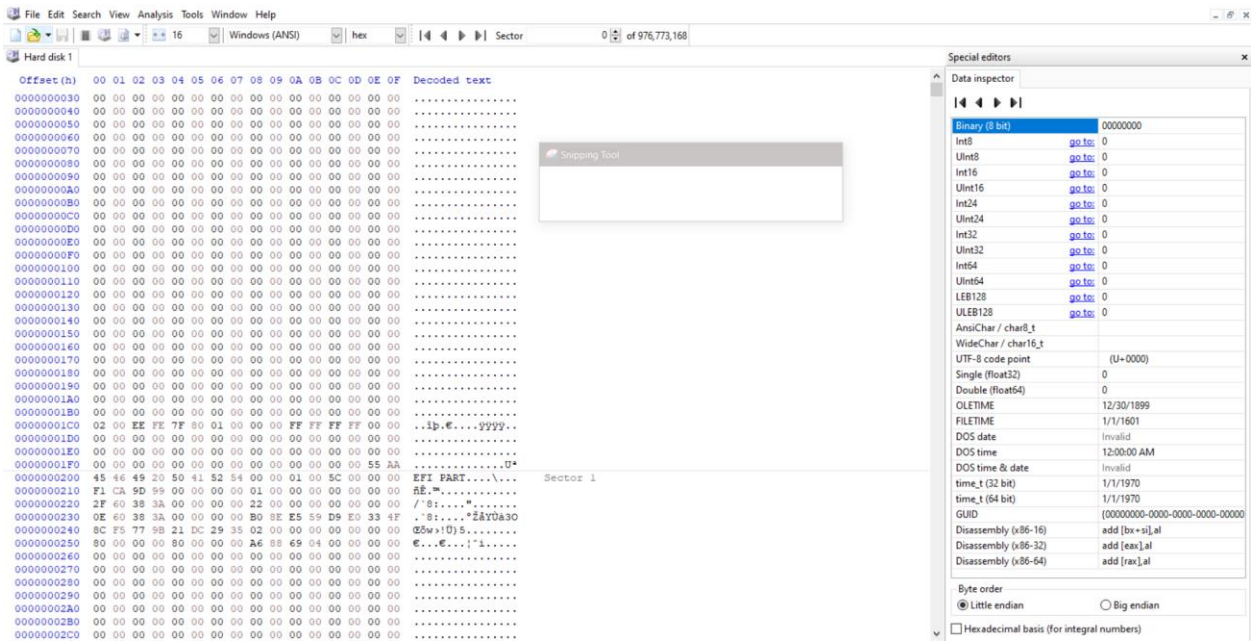
- **Target Disk:** Physical Disk 1 (Basic)
- **Partition Style:** GUID Partition Table (GPT)
- **Forensic Tool:** HxD Hex Editor (Run as Administrator)
- **Verification Tool:** Windows Disk Management



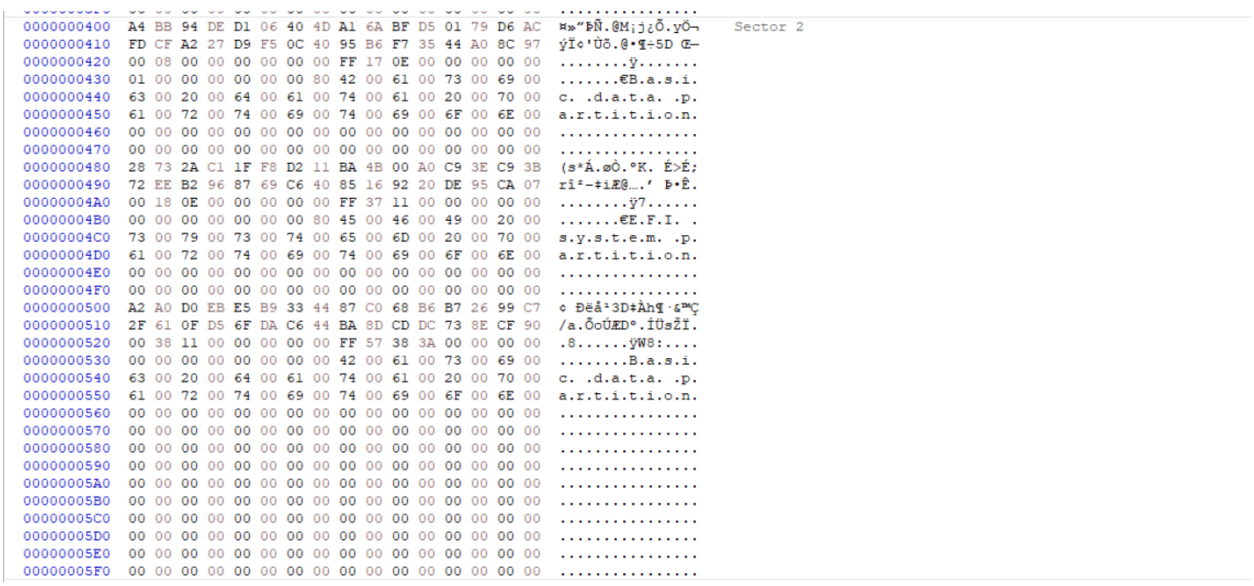
2. Accessing the Partition Table in HxD:

The physical drive was loaded into HxD Editor in read-only mode. I navigated to Sector 2 (Offset 400) to locate the first partition entry. In GPT, each partition entry is 128 bytes long, and the critical boundaries (LBA) are stored in 8-byte sequences.

Sector 1:



Sector 2:

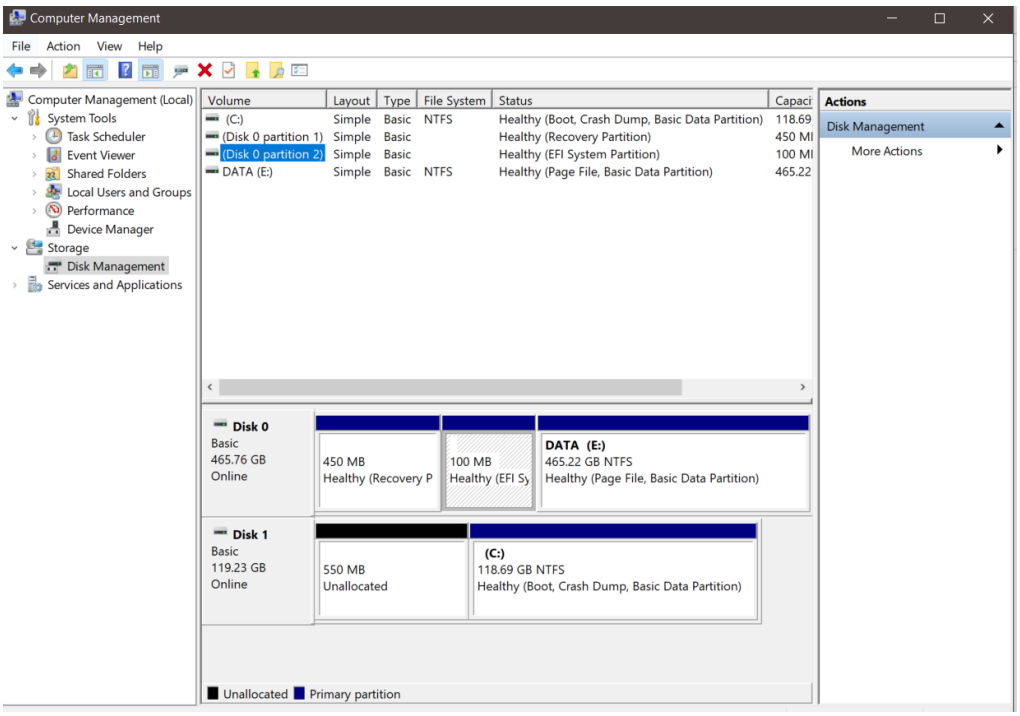


3. Technical Calculation (Partition 1)

To calculate the exact size of the first partition, I extracted the **Starting LBA** and **Ending LBA** from the hex values at Offset `420` and `428`.

A. Data Extraction

- **Starting LBA (at Offset 420):** `00 08 00 00 00 00 00 00`
  - *Conversion (Little Endian to Decimal):* **2,048**
- **Ending LBA (at Offset 428):** `FF 17 0E 00 00 00 00 00`
  - *Conversion (Little Endian to Decimal):* **923,647**



4. Analysis of Physical Disk 0 (465.76 GB Total)

Disk 0 contains three main partitions: a Recovery partition, an EFI system partition, and a large Data partition (E:)

Partition 1: Recovery Partition

Starting LBA: 2048
Ending LBA: 923647

Calculation of Total Sectors:
(923647 - 2048) + 1 = 921600 sectors

Size Calculation in GB:
(921600 * 512) / (1024^3) = 0.43 GB

Result:
The calculated size is approximately 0.43 GB, which matches the 450 MB shown in Disk Management.

Partition 2: EFI System Partition

Starting LBA: 923648
Ending LBA: 1128447

Calculation of Total Sectors:
(1128447 - 923648) + 1 = 204800 sectors

Size Calculation in GB:
(204800 * 512) / (1024^3) = 0.095 GB

Result:
The calculated size is approximately 0.095 GB, which matches the 100 MB shown in Disk Management.

Partition 3: Data (E:)

Starting LBA: 1128448
Ending LBA: 976771071

Calculation of Total Sectors:
(976771071 - 1128448) + 1 = 975642624 sectors

Size Calculation in GB:
(975642624 * 512) / (1024^3) = 465.22 GB

Result:
The calculated size is 465.22 GB, which exactly matches the value displayed in Disk Management.

5. Hard Disk 2:

Sector 1:

```
0000000200  45 46 49 20 50 41 52 54 00 00 01 00 5C 00 00 00  EFI PART....\...    Sector 1
0000000210  48 65 3F E3 00 00 00 00 01 00 00 00 00 00 00 00  He?ã............
0000000220  AF C2 E7 0E 00 00 00 00 22 00 00 00 00 00 00 00  ¯Âç.....".......
0000000230  8E C2 E7 0E 00 00 00 00 86 2E D9 89 BE 12 46 47  ŽÂç.....†.Ù‰¾.FG
0000000240  A1 A4 D8 E3 5A 85 C9 AA 02 00 00 00 00 00 00 00  ¡¤ØãZ…Éª.......
0000000250  80 00 00 00 80 00 00 00 C7 45 D9 76 00 00 00 00  €...€...ÇEÙv....
0000000260  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
0000000270  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
0000000280  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
0000000290  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
00000002A0  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
00000002B0  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
00000002C0  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
00000002D0  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
00000002E0  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
00000002F0  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
0000000300  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
0000000310  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
0000000320  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
0000000330  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
0000000340  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
0000000350  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
0000000360  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
0000000370  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
0000000380  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
0000000390  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
00000003A0  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
00000003B0  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
00000003C0  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
00000003D0  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
00000003E0  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
00000003F0  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
```

Sector 2:

```
0000000400  16 E3 C9 E3 5C 0B B8 4D 81 7D F9 2D F0 02 15 AE  .ãÉã\.¸M.}ù-ð..®    Sector 2
0000000410  FC 97 2A 3F F1 7A 12 47 BD 18 E0 F4 94 30 FE A7  ü—*?ñz.G½.àô”0þ§
0000000420  00 08 00 00 00 00 00 00 FF 87 00 00 00 00 00 00  ........ÿ‡......
0000000430  00 00 00 00 00 00 00 80 4D 00 69 00 63 00 72 00  .......€M.i.c.r.
0000000440  6F 00 73 00 6F 00 66 00 74 00 20 00 72 00 65 00  o.s.o.f.t. .r.e.
0000000450  73 00 65 00 72 00 76 00 65 00 64 00 20 00 70 00  s.e.r.v.e.d. .p.
0000000460  61 00 72 00 74 00 69 00 74 00 69 00 6F 00 6E 00  a.r.t.i.t.i.o.n.
0000000470  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
0000000480  A2 A0 D0 EB E5 B9 33 44 87 C0 68 B6 B7 26 99 C7  ¢ Ðëå¹3D‡Àh¶·&™Ç
0000000490  85 44 38 CF 12 C5 8F 43 97 0A D0 03 7A 95 C0 C0  …D8Ï.Å.C—.Ð.z•ÀÀ
00000004A0  00 B8 11 00 00 00 00 00 FF BF E7 0E 00 00 00 00  .¸......ÿ¿ç.....
00000004B0  00 00 00 00 00 00 00 00 42 00 61 00 73 00 69 00  ........B.a.s.i.
00000004C0  63 00 20 00 64 00 61 00 74 00 61 00 20 00 70 00  c. .d.a.t.a. .p.
00000004D0  61 00 72 00 74 00 69 00 74 00 69 00 6F 00 6E 00  a.r.t.i.t.i.o.n.
00000004E0  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
00000004F0  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
0000000500  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
0000000510  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
0000000520  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
0000000530  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
0000000540  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
0000000550  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
0000000560  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
0000000570  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
0000000580  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
0000000590  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
00000005A0  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
00000005B0  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
00000005C0  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
00000005D0  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
00000005E0  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
00000005F0  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
```

ULEB128	go to:	1D9DFFF
AnsiChar / char8_t		ÿ
WideChar / char16_t		嘿
UTF-8 code point		Invalid code unit
Single (float32)		5.7130782108721E-30
Double (float64)		1.23550497543284E-315
OLETIME		12/30/1899
FILETIME		1/1/1601 12:00:25 AM
DOS date		Invalid
DOS time		Invalid
DOS time & date		Invalid
time_t (32 bit)		12/4/1977 7:36:31 AM
time_t (64 bit)		12/4/1977 7:36:31 AM
GUID		{0EE7BFFF-0000-0000-0000-00000
Disassembly (x86-16)		Invalid

Manual Calculation in GB C: Drive

1. Total Sectors

Formula:
Total Sectors = (Ending LBA - Starting LBA) + 1

Calculation:
(250,068,991 - 1,161,216) + 1 = 248,907,776 sectors

2. Size in Bytes

Formula:
Size in Bytes = Total Sectors × 512

Calculation:
248,907,776 × 512 = 127,440,781,312 bytes

3. Size in GB (Final Result)

Formula:
Size in GB = Size in Bytes / (1024 × 1024 × 1024)

Calculation:
127,440,781,312 / 1,073,741,824 = 118.69 GB

Result:
The size of C: Drive is **118.69 GB**, which matches the size shown in Disk Management.